

PRACTICE REPORT

COMPUTER NETWORKS

Lab 2

COMPUTER NETWORKS

Teacher: **Phan Trung Phát**

Group: **CS4283.Q12.CTTT**

Name	ID
Nguyễn Dương Phúc	24521386

OTHER EVALUATIONS (*):

Content	Result
Average total time to complete the lab ⁽¹⁾	9 hours
Video Link ⁽²⁾ (if any)	
Feedback ⁽³⁾ (if any) + Challenges + Suggestions ...	
Self-Assessment Score ⁽⁴⁾	9/10

(*): Sections (1) and (4) are mandatory.

TASK ASSIGNMENT TABLE:

Members	Task Details
Nguyễn Dương Phúc	All tasks (Work - 100%)

The section below is the detailed report of the group/individual responsible.



FOR INTERNAL CIRCULATION ONLY



Question 1

1. Evidence:

No.	Time	Source	Destination	Protocol	Length	Info
537	2.855377	192.168.228.54	192.168.228.86	HTTP	496	GET /24521395.html HTTP/1.1
540	2.857577	192.168.228.86	192.168.228.54	HTTP	634	HTTP/1.1 200 OK (text/html)

Figure 1. HTTP version and server version

2. Explanation:

The browser is using HTTP version 1.1 (496 GET /24521395.html HTTP /1.1)

The server is using HTTP version 1.1 (634 HTTP/1.1 200 OK (text/html)).

Question 2

1. Evidence:

No.	Time	Source	Destination	Protocol	Length	Info
537	2.855377	192.168.228.54	192.168.228.86	HTTP	496	GET /24521395.html HTTP/1.1

Figure 2. IP addresses of client and server

2. Explanation:

My computer's IP address: **192.168.228.54** (Source)

The web server's IP address: **192.168.228.86** (Destination).

Question 3

1. Evidence:

No.	Time	Source	Destination	Protocol	Length	Info
537	2.855377	192.168.228.54	192.168.228.86	HTTP	496	GET /24521395.html HTTP/1.1
540	2.857577	192.168.228.86	192.168.228.54	HTTP	634	HTTP/1.1 200 OK (text/html)

Figure 3. HTTP response status code

2. Explanation:

The status code: **200 OK**.



Question 4

1. Evidence:

```
▼ Hypertext Transfer Protocol
  ▶ HTTP/1.1 200 OK\r\n
    Content-Type: text/html\r\n
    Last-Modified: Mon, 13 Oct 2025 08:48:17 GMT\r\n
    Accept-Ranges: bytes\r\n
    ETag: "be36741e1e3cdc1:0"\r\n
    Server: Microsoft-IIS/10.0\r\n
    Date: Mon, 13 Oct 2025 09:03:51 GMT\r\n
  ▶ Content-Length: 355\r\n
    \r\n
    [Request in frame: 537]
    [Time since request: 0.002200000 seconds]
    [Request URI: /24521395.html]
    [Full request URI: http://192.168.228.86/24521395.html]
    File Data: 355 bytes
```

Figure 4. Content length returned by server

2. Explanation:

The server returned **355 bytes of content** to the browser.

Question 5

1. Evidence:

```
▼ Hypertext Transfer Protocol
  ▶ HTTP/1.1 200 OK\r\n
    Content-Type: text/html\r\n
    Last-Modified: Mon, 13 Oct 2025 08:48:17 GMT\r\n
    Accept-Ranges: bytes\r\n
    ETag: "be36741e1e3cdc1:0"\r\n
    Server: Microsoft-IIS/10.0\r\n
    Date: Mon, 13 Oct 2025 09:03:51 GMT\r\n
  ▶ Content-Length: 355\r\n
    \r\n
    [Request in frame: 537]
    [Time since request: 0.002200000 seconds]
    [Request URI: /24521395.html]
    [Full request URI: http://192.168.228.86/24521395.html]
    File Data: 355 bytes
  ▶ Line-based text data: text/html (12 lines)
```

Figure 5. First HTTP GET contents (IF-MODIFIED-SINCE)

2. Explanation:

No, the first HTTP GET request does not contain the line "IF-MODIFIED-SINCE".



Question 6

1. Evidence:

```
▼ Line-based text data: text/html (12 lines)
<!DOCTYPE html>\r\n
<html>\r\n
<head>\r\n
  <meta charset="UTF-8">\r\n
  <title>Thực hành Mạng máy tính - 2022</title>\r\n
</head>\r\n
<body>\r\n
  <center></c
  <center><h1>MS5V: 24521395</h1></center>\r\n
  <center><h2> Họ và tên: Phan Ngoc Phuc</h2></center>\r\n
</body>\r\n
</html>\r\n
```

Figure 6. Server response content (HTML file?)

2. Explanation:

Yes, because Wireshark includes a section “Line-based text data” showing what the server sent back to the browser — the HTML code of the page, as shown in the figure above.

Question 7

1. Evidence:

```
▼ Hypertext Transfer Protocol
  ▶ GET /24521395.html HTTP/1.1\r\n
    Host: 192.168.228.86\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,i
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    If-None-Match: "be36741e1e3cdc1:0"\r\n
    If-Modified-Since: Mon, 13 Oct 2025 08:48:17 GMT\r\n
    \r\n
    [Response in frame: 1059]
    [Full request URI: http://192.168.228.86/24521395.html]
```

Figure 7. Second HTTP GET (IF-MODIFIED-SINCE value)

2. Explanation:

Yes, the second HTTP GET contains the line “**If-Modified-Since: Mon, 13 Oct 2025 08:48:17 GMT**”.



Question 8

1. Evidence:

```
▼ Hypertext Transfer Protocol
  ▶ HTTP/1.1 304 Not Modified\r\n
    Accept-Ranges: bytes\r\n
```

Figure 8. HTTP status code of second GET

2. Explanation:

The HTTP status code returned from the server: **304 Not Modified (HTTP/1.1 304 Not Modified)**.

It means: the requested file **has not been changed** since the last request

The server **does not resend** the file contents

Because the browser **already has** a cached copy of the HTML file **from the first HTTP GET** and **loads** the page **from its cache** instead.

Question 9

1. Evidence:

No.	Time	Source	Destination	Protocol	Length	Info
537	2.855377	192.168.228.54	192.168.228.86	HTTP	496	GET /24521395.html HTTP/1.1
540	2.857577	192.168.228.86	192.168.228.54	HTTP	634	HTTP/1.1 200 OK (text/html)
552	2.873082	192.168.228.54	10.204.2.4	HTTP	470	GET /sites/default/files/photos/large/202110/kimg0816.jpg HTTP/1.1
954	2.880921	10.204.2.4	192.168.228.54	HTTP	636	HTTP/1.1 200 OK (JPEG JFIF image)
956	2.885242	192.168.228.54	192.168.228.86	HTTP	439	GET /favicon.ico HTTP/1.1
957	2.888081	192.168.228.86	192.168.228.54	HTTP	1437	HTTP/1.1 404 Not Found (text/html)
1056	4.899716	192.168.228.54	192.168.228.86	HTTP	608	GET /24521395.html HTTP/1.1
1059	4.903567	192.168.228.86	192.168.228.54	HTTP	197	HTTP/1.1 304 Not Modified

Figure 9. Browser HTTP GETs and IP addresses

2. Explanation:

The browser sent 4 HTTP GET requests:

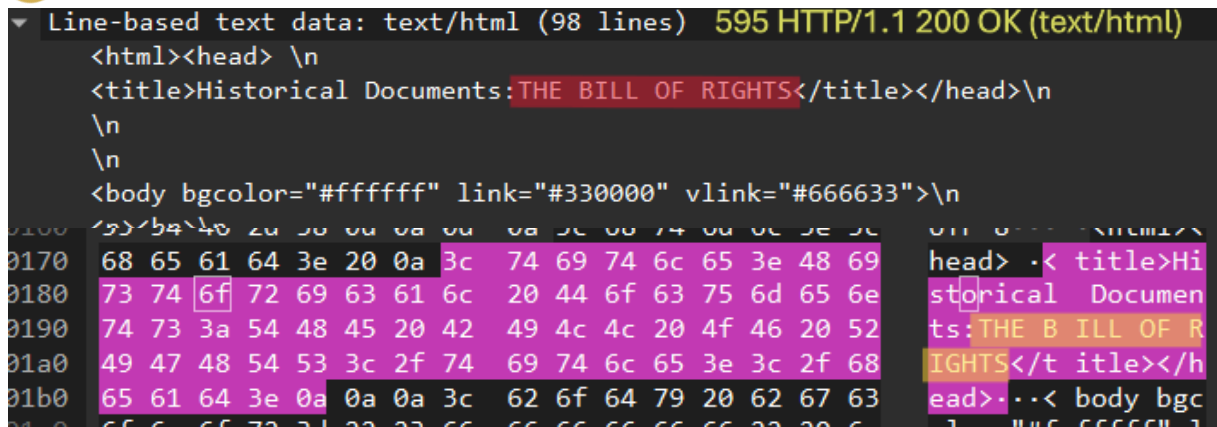
- 496 GET /24521395.html → 192.168.228.86
- 470 GET /sites/default/files/photos/large/202110/kimg0816.jpg → 10.204.2.4
- 439 GET /favicon.ico → 192.168.228.86
- 608 GET /24521395.html → 192.168.228.86

Question 10

1. Evidence:

No.	Time	Source	Destination	Protocol	Length	Info
2183	2.961691	10.0.123.222	128.119.245.12	HTTP	549	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
2263	3.229083	128.119.245.12	10.0.123.222	HTTP	595	HTTP/1.1 200 OK (text/html)
2288	3.275209	10.0.123.222	128.119.245.12	HTTP	495	GET /favicon.ico HTTP/1.1
2474	3.537835	128.119.245.12	10.0.123.222	HTTP	538	HTTP/1.1 404 Not Found (text/html)

Figure 20. Browser HTTP GET requests



```

Line-based text data: text/html (98 lines) 595 HTTP/1.1 200 OK (text/html)
<html><head> \n
<title>Historical Documents: THE BILL OF RIGHTS</title></head>\n
\n
\n
<body bgcolor="#ffffff" link="#330000" vlink="#666633">\n
head> .< title>Hi
istorical Documen
ts: THE B ILL OF R
IGHTS</t itle></h
ead>...< body bgc

```

Figure 31. Response packet containing "THE BILL OF RIGHTS"

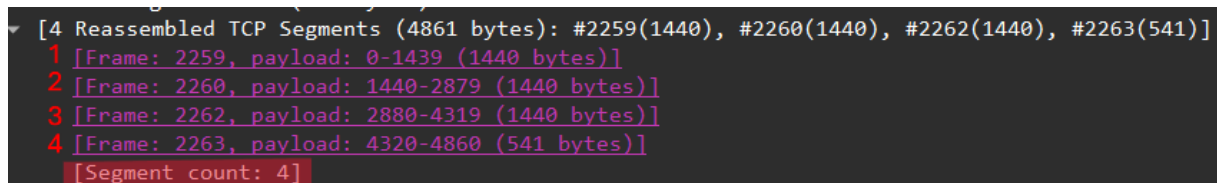
2. Explanation:

The browser sent 2 HTTP GET requests: frame 2183 (549 GET) and frame 2288 (495 GET).

The line "THE BILL OF RIGHTS" is contained in the first response frame 2263 (595 HTTP/1.1 200 OK, text/html).

Question 11

1. Evidence:



```

[4 Reassembled TCP Segments (4861 bytes): #2259(1440), #2260(1440), #2262(1440), #2263(541)]
1 [Frame: 2259, payload: 0-1439 (1440 bytes)]
2 [Frame: 2260, payload: 1440-2879 (1440 bytes)]
3 [Frame: 2262, payload: 2880-4319 (1440 bytes)]
4 [Frame: 2263, payload: 4320-4860 (541 bytes)]
[Segment count: 4]

```

Figure 42. TCP segments carrying "The Bill of Rights"

2. Explanation:

There are 4 TCP segments to carry the entire HTTP response and the content of "The Bill of Rights".

Question 12

1. Evidence:

No.	Time	Source	Destination	Protocol	Length	Info
757	3.793331	10.0.123.222	128.119.245.12	HTTP	587	GET /wireshark-labs/protected_pages/HTTP-wireshark-%20file5.html HTTP/1.1
804	4.066386	128.119.245.12	10.0.123.222	HTTP	771	HTTP/1.1 401 Unauthorized (text/html)

Figure 53. Status code of first HTTP GET

2. Explanation:

The status code in the first HTTP GET: **401 Unauthorized**.

This means the page cannot be accessed until the correct username and password are provided.



Question 13

1. Evidence:

```
▼ Hypertext Transfer Protocol
  ▶ GET /wireshark-labs/protected_pages/HTTP-wireshark-%20file5.html HTTP/
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    ▶ Authorization: Basic d2lyZXNoYXJrLXN0dWR1bnRzOm5ldHdvcm5=\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537
```

Figure 64. Authorization field in second HTTP GET

2. Explanation:

The new field that appears in the second HTTP GET: the **Authorization** field.

END